

Linux Service Discovery and Enumeration

Nmap, Netcat, Banner Analysis, and Web-Service Review

ETHICAL HACKING

CONTROLLED LAB

EVIDENCE BASED

An authorized Linux enumeration exercise that resolves a lab target, inventories TCP services, investigates ambiguous ports, confirms an intentionally exposed bind shell, and reviews a vulnerable web surface.

Authorization and safety boundary: This report documents an instructor-provided, isolated training exercise. No real users, public targets, production systems, or real credentials were used. Commands use placeholders for portfolio safety. Original screenshots remain unchanged and may show transient private lab addresses.

ASSESSMENT TYPE	Authorized training-lab exercise
OBJECTIVE	Build a defensible service inventory and identify exposures requiring deeper validation or immediate containment.
ENVIRONMENT	Instructor-provided Kali Linux and intentionally vulnerable Linux target on an isolated RFC1918 network.
EVIDENCE	5 original screenshots retained unchanged and presented in execution order.

Professional Skills Demonstrated

- Target validation
- Nmap service detection
- Netcat banner analysis
- Unexpected service investigation
- Web enumeration
- Risk-based remediation

Assessment Workflow and Evidence

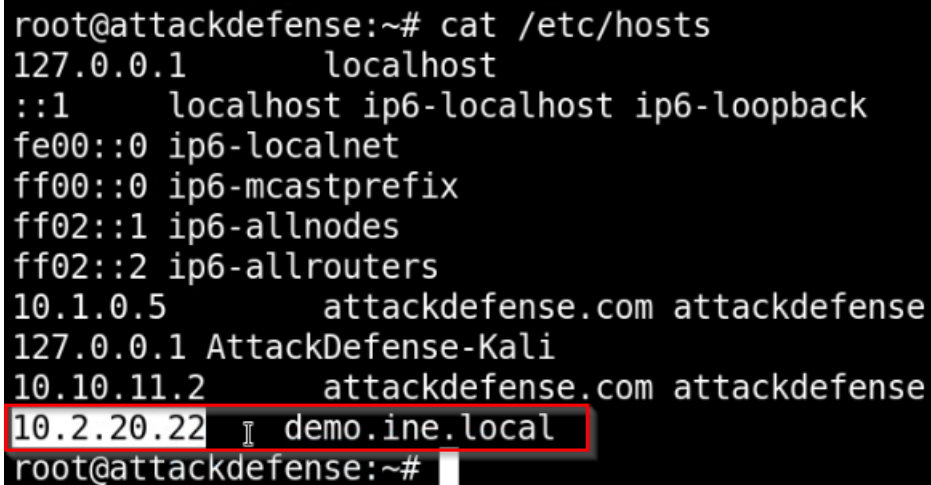
1. Resolve and Confirm the Target

Objective: Identify the lab-only target before scanning.

The local hosts file was reviewed to confirm the instructor-provided target mapping. Commands in this public report use TARGET_IP because the screenshot address was transient and specific to the lab session.

Lab Commands

```
cat /etc/hosts  
  
ping -c 2 demo.ine.local
```



```
root@attackdefense:~# cat /etc/hosts  
127.0.0.1      localhost  
::1           localhost ip6-localhost ip6-loopback  
fe00::0       ip6-localnet  
ff00::0       ip6-mcastprefix  
ff02::1       ip6-allnodes  
ff02::2       ip6-allrouters  
10.1.0.5       attackdefense.com attackdefense  
127.0.0.1      AttackDefense-Kali  
10.10.11.2     attackdefense.com attackdefense  
10.2.20.22     demo.ine.local  
root@attackdefense:~#
```

Evidence 01 - Lab target resolution using the local hosts file. Original lab screenshot retained unchanged.

Analyst observation: Target confirmation reduces the risk of scanning the wrong system and establishes a reproducible starting point.

2. Build the Service Inventory

Objective: Identify listening TCP services and their reported versions.

A service-detection scan across the first 10,000 TCP ports produced a broad inventory. The result was used for prioritization rather than treated as proof of vulnerability.

Lab Commands

```
nmap -sV -p1-10000 TARGET_IP -oA evidence/linux-service-inventory
```

PORT	STATE	SERVICE	VERSION
21/tcp	open	ftp	vsftpd 2.3.4
22/tcp	open	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.2 (Ubuntu Linux; protocol 2.0)
23/tcp	open	telnet	Linux telnetd
25/tcp	open	smtp	Postfix smtpd
51/tcp	open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
80/tcp	open	http	Apache httpd/2.2.8 ((Ubuntu) DAV/2)
111/tcp	open	rpcbind	2 (RPC #100000)
139/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp	open	exec?	
513/tcp	open	login?	
514/tcp	open	tcpwrapped	
1099/tcp	open	java-rmi	GNU Classpath grmiregistry
1524/tcp	open	ingreslock?	
3306/tcp	open	mysql	MySQL 5.0.51a-3ubuntu5
3632/tcp	open	distccd	distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp	open	postgresql	PostgreSQL DB 8.3.0 - 8.3.7
6667/tcp	open	irc	UnrealIRCd
6697/tcp	open	irc	UnrealIRCd
8009/tcp	open	ajp13	Apache Jserv (Protocol v1.3)
8180/tcp	open	http	Apache Tomcat/Coyote JSP engine 1.1
8787/tcp	open	drb	Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drb)

Evidence 02 - Nmap service and version inventory. Original lab screenshot retained unchanged.

Analyst observation: The scan exposed several legacy services and two ports requiring manual validation.

3. Validate Ambiguous Ports

Objective: Use a simple TCP client to observe application behavior on ports without reliable banners.

Netcat produced no useful banner on one service, while the second connection returned an interactive shell. This was an intentionally vulnerable lab condition and was not used to modify the host.

Lab Commands

```
nc -nv TARGET_IP 512
nc -nv TARGET_IP 1524
```

```
root@attackdefense:~# nc -nv 10.2.20.22 512
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: Connected to 10.2.20.22:512.

ls
^C
root@attackdefense:~#
```

Evidence 03 - Manual connection tests for ambiguous TCP services. Original lab screenshot retained unchanged.

```
root@attackdefense:~# nc -nv 10.2.20.22 1524
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: Connected to 10.2.20.22:1524.
root@d1d6a9361621:/# ls
ls
bin
boot
cdrom
core
dev
etc
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
```

Evidence 04 - Manual connection tests for ambiguous TCP services. Original lab screenshot retained unchanged.

Analyst observation: A listening port with no banner still requires investigation; an unauthenticated bind shell is a critical remote-access exposure.

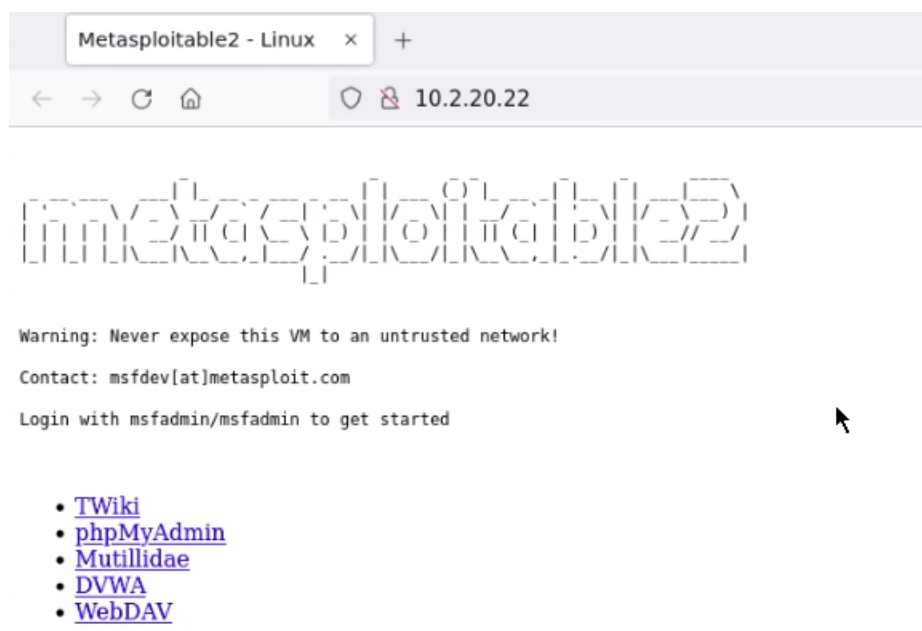
4. Review the Web Surface

Objective: Inspect the HTTP landing page and visible application paths.

The web page exposed multiple intentionally vulnerable applications and a WebDAV path. The observation created a prioritized follow-up list without expanding the public report into additional exploitation steps.

Lab Commands

```
curl -I http://TARGET_IP/
nmap -sV -p80 --script http-title,http-methods TARGET_IP
```



Evidence 05 - Vulnerable training web surface and exposed application paths. Original lab screenshot retained unchanged.

Analyst observation: Visible test applications and WebDAV increase attack surface and should be removed, isolated, or explicitly required and hardened.

Findings and Risk

Severity	Finding	Evidence-based impact
CRITICAL	Unauthenticated bind shell	A remote client obtained an interactive shell through an exposed TCP service in the intentionally vulnerable lab.
HIGH	Legacy and unnecessary services	The scan identified multiple old services that expand exposure and complicate monitoring.
HIGH	Vulnerable web applications	The HTTP service advertised deliberately vulnerable applications and a potentially misconfigured WebDAV path.
MEDIUM	Incomplete service identification	A port without a banner requires process, socket, and packet-level correlation before classification.

Defensive Recommendations

- Remove bind shells, remote debug listeners, and every service without a documented business requirement.
- Restrict management access with host firewalls, network ACLs, and a dedicated administrative segment.
- Patch or replace legacy daemons and disable insecure protocols.
- Disable WebDAV unless required; otherwise enforce authentication, least privilege, and limited methods.
- Correlate scan results with host commands such as ss, lsof, systemctl, package inventory, and process ownership.
- Create detections for new listeners, unusual parent processes, and inbound connections to nonstandard ports.

Reflection and Next Step

The exercise reinforced that enumeration is evidence collection, not automatic vulnerability proof. The bind-shell result was decisive because service behavior was directly observed, while other ports still required host or protocol evidence before drawing conclusions.

Evidence note: The screenshots are original lab captures retained unchanged. The surrounding narrative was rewritten for clarity, public safety, and recruiter review. Training-provider names, marks, and course materials remain the property of their respective owners; this portfolio is not an endorsement or affiliation claim.